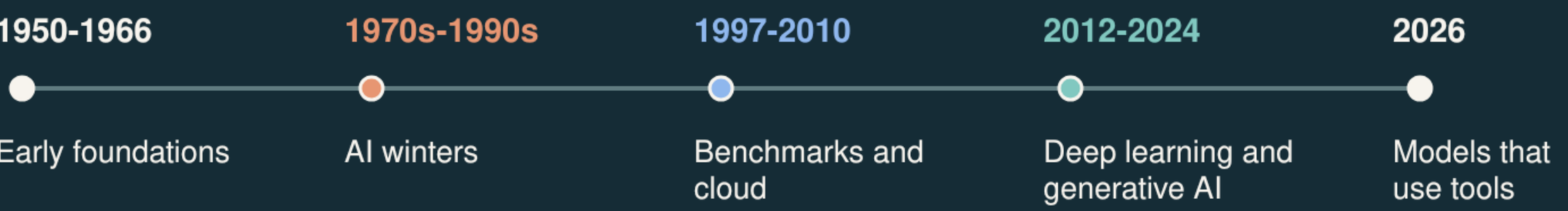
Artificial intelligence
A visual history

1950-2026



Yu-Hsuan Tseng
Selected milestones through October 18, 2026

1950-1966: Early foundations

Researchers began testing how machines could learn and converse.

1950

The imitation game

Turing proposes a conversational test for machine intelligence, framing a lasting question about how to evaluate intelligent behavior.

Alan Turing

[1]

1956

The Dartmouth workshop

The summer research project helps establish AI as a named field. Its organizers wrote the proposal in 1955.

John McCarthy, Marvin Minsky
Nathaniel Rochester
Claude Shannon

[2]

1958

The perceptron

Rosenblatt demonstrates a system that learns to classify patterns, laying an early foundation for neural network research.

Frank Rosenblatt
Cornell Aeronautical Laboratory

[3]

1966

ELIZA

A program uses pattern matching to imitate conversation, showing how simple rules can create an impression of understanding.

Joseph Weizenbaum
MIT

[4]

Chronological sequence. Spacing is not proportional to elapsed time.

1973-1986: Setbacks and renewed interest

The first AI winter reduced support, while useful applications and neural learning continued.

1973

The Lighthill report

Lighthill questions progress in general AI. His review weakens confidence in UK research, especially in robotics and language processing.

James Lighthill
UK Science Research Council

[5]

c. 1974-1980

First AI winter

Cause: methods struggled beyond small demonstrations.
Impact: governments cut funding and research programs contracted.

Government funders
University research groups

[5] [6]

1980

R1 / XCON

A rule-based expert system configures computer orders, showing that specialized AI can solve a practical business problem.

John McDermott
Carnegie Mellon and DEC

[7]

1986

Backpropagation

Networks learn internal representations by propagating errors backward. The paper renews interest in training multilayer networks.

David Rumelhart
Geoffrey Hinton, Ronald Williams

[8]

Winter dates are approximate. Funding and research activity varied by country and specialty.

1987-2010: A second winter and wider access

Commercial disappointment coexisted with advances in specialized systems and infrastructure.

c. 1987-1995

Second AI winter

Cause: brittle, costly systems and hardware competition.
Impact: investment cooled and research priorities shifted.

Expert system companies
Specialized hardware vendors

[6]

1997

Deep Blue wins

IBM's chess computer beats world champion Garry Kasparov in a match, demonstrating the power of specialized search and hardware.

Feng-hsiung Hsu, Murray Campbell
and the IBM team

[9]

2006

EC2 enters beta

Amazon offers computing capacity on demand, reducing the need to buy servers before running computational workloads.

Amazon Web Services
Cloud infrastructure

[10]

2010

Cloud GPUs

AWS introduces rental access to NVIDIA Tesla GPUs, making powerful parallel processing available through the cloud.

Amazon Web Services
NVIDIA

[11]

Winter dates are approximate. Research continued during both downturns. Timeline spacing is schematic.

2012-2017: Deep learning changes the field

Large datasets and faster hardware help neural networks deliver stronger results.

2012

AlexNet

A deep convolutional network achieves a major ImageNet breakthrough. GPU computation makes training the large network practical.

Alex Krizhevsky, Ilya Sutskever, Geoffrey Hinton

[12]

2014

GANs

Generative adversarial networks pair a generator and discriminator, advancing methods for creating samples resembling training data.

Ian Goodfellow and collaborators Including Yoshua Bengio

[13]

2016

AlphaGo

DeepMind combines neural networks with search and reinforcement learning. In March, AlphaGo defeats Lee Sedol 4-1.

David Silver, Demis Hassabis and the DeepMind team

[14] [15]

2017

The Transformer

An attention-based architecture improves parallel training for language tasks and helps underpin later large language models.

Ashish Vaswani and collaborators Google research teams

[16]

Chronological sequence. Spacing is not proportional to elapsed time.

2020-2024: Generative AI reaches more users

AI expands into language interfaces, scientific research and multimodal interaction.

2020

GPT-3

A large language model performs many tasks from a few examples in the prompt, demonstrating the benefits of scaling language models.

Tom Brown and collaborators
OpenAI

[17]

2021

AlphaFold 2 paper

Published after its 2020 CASP14 breakthrough, the system predicts protein structures with strong accuracy, supporting biological research.

John Jumper, Demis Hassabis and the DeepMind team

[18]

2022

ChatGPT

OpenAI launches a conversational research preview trained with human feedback, bringing generative language tools to more users.

OpenAI
Launch: November 30

[19]

2024

GPT-4o

A multimodal model combines text, image and audio capabilities, expanding natural interaction and assistive applications.

OpenAI
Announcement: May 13

[20]

Chronological sequence. Spacing is not proportional to elapsed time.

2026: Models increasingly use tools

Two examples of the shift toward agents that carry out work across multiple steps.

May 19

Gemini 3.5 Flash

Google introduces a model for coding and agent workflows, with access through developer tools and enterprise platforms.

Google DeepMind
Koray Kavukcuoglu, Jeff Dean and colleagues

[21]

September 29

GPT-6.1 Sol at DevDay

OpenAI announces an upgraded model for coding, computer use and professional tasks, alongside reusable cloud development environments.

OpenAI
Model and cloud product teams

[22]

Selected product examples. Vendor announcements document release dates and intended uses.

Chronological sequence. Spacing is not proportional to elapsed time.

What enabled the recent advances?

The historical pattern depends on both better models and the resources to build and run them.

Deep learning

Networks learn useful representations from data. AlexNet demonstrated image recognition gains, and Transformers improved parallel training for language tasks.

[12] [16]

Cloud computing

Rental compute and GPUs broaden access to infrastructure. Cloud environments also help teams run and deliver AI applications without first owning every server.

[10] [11] [22]

Task success and general intelligence

Strong results in chess, Go or a language benchmark establish specific capabilities. AGI refers to broader general capability, with definitions varying across organizations. [23, 24]

Chronological sequence. Spacing is not proportional to elapsed time.

Research and AI assistance

An original timeline layout, supported by linked sources.

Tools used

ChatGPT assisted with milestone research, web source checking, text drafting and slide design. The timeline uses original wording and an original arrangement of editable PowerPoint objects.

Source choices and scope

Original papers, institutional histories and dated product announcements support the timeline. Winter dates are approximate. The Our World in Data reading is credited to Max Roser (2022). [23]

Research checked through October 18, 2026. Numbered citations link to the references on the following slides.

References 1 of 3

Reference numbers match the timeline. Each title links to the source.

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